		$g = \dfrac{\ldots}{(t_2^2 - t_1^2)\sin\theta}$
4	**A**	Rate of change of momentum $= \dfrac{\Delta(mv)}{t}$ $= v_{rel}\dfrac{\Delta m}{t}$ $= v_{rel}\dfrac{(\rho \times \text{vol of water expelled})}{t}$ $= v_{rel}\dfrac{(\rho \times \pi r^2 h)}{t}$ $= v\rho\pi r^2\dfrac{h}{t}$ $= v\rho\pi r^2 v$ $= \rho\pi r^2 v^2$ $= 1000 \times \pi \times 0.012^2 \times 10^2$ $= 45\text{ N}$

5	C	Assume Charlie throws the ball to the right and take rightwards as positive.